

$$* \langle \underline{2}y_n, y_m \rangle = 2 \langle y_n, y_m \rangle$$

$$* \langle A_n y_n, A_m y_m \rangle = A_n A_m \langle y_n, y_m \rangle \stackrel{!}{=} \delta_{nm}$$

$$= \alpha_n \delta_{nm}$$

$$\text{(i)} \quad n \neq m : = 0$$

$$\text{(ii)} \quad n = m : A_n A_n \alpha_n = 1$$

$$A_n^2 \alpha_n = 1 \Rightarrow A_n = \frac{1}{\sqrt{\alpha_n}}$$

$$\rightarrow \langle y_n, y_m \rangle = \alpha_n \delta_{nm}$$

$$n \neq m : 0$$

$$n = m : \langle y_n, y_n \rangle = \alpha_n$$

$$A_n = \frac{1}{\sqrt{\langle y_n, y_n \rangle}}$$

$$* \text{Normalized basis vectors: } \hat{y}_n(x) \equiv A_n y_n$$

$$\hat{y}_n(x) = \frac{y_n}{\sqrt{\langle y_n, y_n \rangle}}$$

$$* \text{Recall, : } \mathbb{R}^3 \quad \vec{x} : \hat{x} \equiv \frac{\vec{x}}{\|\vec{x}\|} \quad *$$

$$\|y_n\| = \sqrt{\langle y_n, y_n \rangle}$$

$$\|\vec{x}\|^2 = \vec{x} \cdot \vec{x} = \langle x, x \rangle$$

$$\|\vec{x}\| = \sqrt{\langle x, x \rangle}$$

PS4 Q6 : $t \geq 0$

$$\text{B.C's: } u(0, t) = 0, \quad u(2, t) = 0$$

$$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$$

$$x=0$$

$$x=2$$

ex: $u(0, y) = 0, \quad u(3\pi, y) = 0$

$u(x, 1) = 0, \quad u(x, \pi) = 0$

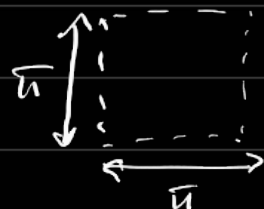
$$\begin{cases} 0 < x < 3\pi \\ 1 < y < \pi \end{cases}$$

$$\begin{cases} 0 < x < 3\pi \\ 1 < y < \pi \end{cases}$$

* L: Generally "size" of Boundary

Q7: $L = \pi$

Q6: $L = 2$



Real "steady state"

* $u(x, t) = \underbrace{u(x)}_{\text{steady state}} + \underbrace{v(x, t)}_{\text{transient}}$

only of x: so I.C.'s Don't apply

* Time dependent B.C.'s:

$$u(x, t): \begin{cases} u(0, t) = 0 \\ u(1, t) = e^{-t} \end{cases}$$

before $u(1, t) = 0$, or $= 1$,

13th: Quiz: * $u_{tt} = u_{xx} + \cos(x) \sin(t)$ *

* $u_x(0, t) = u_x(\pi, t) = 0$ (B.C's)

$u(x, 0) = u_t(x, 0) = 0$

$x \in (0, \pi), \quad t \geq 0$

$u_n = \cos(nx)$

Not Time dep. (B.C's)

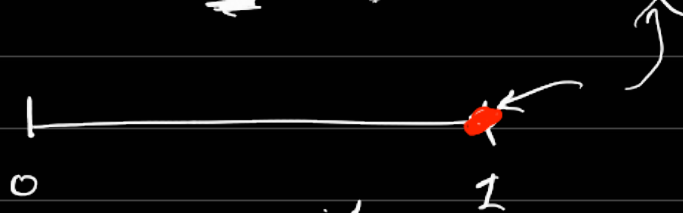
(PSS) 16) (c)

* Note: $(x, y) \Rightarrow$ The entire problem is the ind.

(ex) Nov 13: $u_t = u_{xx}$

B.C's: $u(0, t) = 0, \quad u(1, t) = e^{-t}$

I.C.: $u(x, 0) = x$



ex: $y'' = e^t$ / $y'' = e^{-t}$
 $y = e^t$

* $u(x, t) = \psi(x, t) + v(x, t) \leftarrow$

B.C. + I.C.

I.C.: $u(x, 0) = \psi(x, 0) + \underbrace{v(x, 0)}_{\text{Homogeneous}} = x$

$\begin{cases} \psi(x, 0) = x \\ \psi(1, t) = e^{-t} = \psi(1, t) + \underbrace{v(1, t)} \end{cases}$
 $\Rightarrow \psi(1, t) = e^{-t}$

$\psi(x, t) = x e^{-t}$

* y_n : $y_n = ? \leftarrow$ one solⁿ for a given n
 $y_0, y_1, \dots$

* Superpos: two (or more) solⁿ y_1, y_2
 then

$y_1 + y_2$
 is also a sol.

$$u(x, t) = \sum_{n=0}^{\infty} b_n^{(t)} y_n$$

(PSS) (6) $\nabla^2 u = f(x, y)$, $f(x, y) = 1$

Take function a "u" and "v" and "w" and "x" and "y" and "z" and "t" and "s" and "r" and "q" and "p" and "o" and "n" and "m" and "l" and "k" and "j" and "i" and "h" and "g" and "f" and "e" and "d" and "c" and "b" and "a"

technically: we can always write $u(x,t)$ for

both examples earlier.

$$\Rightarrow \text{"Magic } u" : u = u(x) + v(x,t)$$

Based on the problem set up is where things change:

$$u(0,t) = 0, \quad u(1,t) = \underline{1}$$

$$u(1,t) = 1 \quad \leftarrow \text{Doesn't depend on time}$$

$\Rightarrow$ Steady state $\Rightarrow$ completely don't write t

$$(6) \left(\begin{array}{l} u(x,y) = u(y) + v(x,y) \\ \rightarrow \text{wrote } \cancel{u(1,y) = 1} \quad u(x,1) = 1 \\ \rightarrow u(x,y) = \cancel{u(x,y)} + \underline{v(x,y)} \end{array} \right)$$

extra: $\nabla^2 u = \underline{f} = \underbrace{\nabla^2 u}_{\text{"Absorb"}} + \nabla^2 v$

$$* \nabla^2 u = \underline{f} * \quad f(x,y) = 1$$

$$\nabla^2 v = 0$$

$$\rightarrow \text{Solve: } u_{xx} + u_{yy} = 1$$

* PS5 [6] There two inhomogen. that we want to get rid of

(i) In the PDE: $\nabla^2 u = f$ *

(ii) In the B.C. (c): $u(x,1) = 1$ *

$$x \in [0,1]$$

$$y \in [0,1]$$

